

[DC] Mixed Reality Interaction for Mobile Knowledge Work

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ABSTRACT

Knowledge workers typically work on some kind of computer and other than an internet connection, they rarely need access to specific work environments or devices. This makes it feasible for them to also work in different environments or in mobile settings, like public transportation. In such spaces comfort and productivity could be decreased due to hardware limitations like small screen sizes or input devices and environmental clutter. Mixed reality (MR) has the potential to solve such issues. It can provide the user with additional display space that can even include the third dimension. It can open up new possibilities for interacting with virtual content using gestures or spatially tracked devices. And it can allow the users to modify the work environment according to their personal preferences. This doctoral thesis aims at exploring the challenges of using MR for mobile knowledge work and how to effectively support knowledge worker tasks through appropriate interaction techniques.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION AND RELATED WORK

Devices like laptops or tablets that are commonly used by knowledge workers in mobile scenarios, have certain limitations. Small screen sizes or missing input devices such as a mouse might influence the productivity. Mixed reality provides new possibilities to enhance and facilitate knowledge work [6]. A core advantage of using mixed reality for knowledge work is the large output space that allows the user to create an optimal work environment regardless of the real physical surroundings. The user can freely choose the size, number and placement of virtual screens. This is especially interesting for knowledge workers that frequently change their workplace or work on-the-go, for example while commuting or traveling [10]. The three-dimensional output space also enables new ways of presenting information and interacting with it. This, however, requires thorough investigation of interaction techniques to optimally support knowledge worker tasks.

Beyond a large output space, the possibility of easily adapting the virtual environment has further advantages. Using the optimal virtual surrounding could be used to improve the work experience by reducing stress and improving flow [11]. MR is especially suited to block out visual distractions either by completely replacing the real surroundings with a virtual environment or just using virtual separators [8]. However, replacing the physical environment completely creates new challenges. In many scenarios it is not practical to lose touch to the physical surrounding completely. Therefore, parts of it might need to be included [9] such as keyboards or other important objects and also other people like colleagues or other passengers. This means such objects need to be identified and visualized appropriately. In mobile environments such as public transportation this aspect can be very safety critical. Another concern is that of how people outside of the virtual environment react to the user and which possibilities they have to interact with them [4].

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However, MR can also actively enhance the collaboration of multiple people by enabling remote workers to meet in a shared virtual space. Recent work suggests that this can increase productivity in collaborative tasks in comparison to common tools like screen sharing [3].

All these potential benefits imply that it is usable to work in a virtual environment regularly for enhanced periods of time. So far, the longest experiment exploring the use of VR HMDs for some predefined knowledge worker tasks lasted eight hours [7, 13] reporting a higher mental and visual fatigue. However, considering users might need some time to get accustomed to the new environment this experiment is still rather short. In addition, the artificial tasks used could influence the results and it is not sure that these findings hold for real-life scenarios. Therefore, this should be investigated further.

This doctoral thesis focuses on interaction techniques for knowledge work in MR. Current techniques used in MR such as gestures or controllers might not be precise enough for some common knowledge worker tasks. On the other hand, techniques established in conventional knowledge work might not be as usable in MR. Additionally such techniques are designed for 2D interactions and might not support the 3D interaction space of MR. Therefore, the goal of this doctoral thesis is to explore interaction techniques for knowledge work in MR, to identify limitations and then propose and evaluate possible solutions. Specifically, I am looking at interaction techniques for mobile spaces and for supporting collaborative tasks. In addition, I explore the long-term use to validate the feasibility of regularly using MR for work.

2 CURRENT PROGRESS

A main advantage of using MR for mobile knowledge work is the large virtual display space which can be used to overcome the limitations of small physical screens. For example, a large number of virtual screens can be placed around the user. This, however, opens up the question of how to efficiently interact with the virtual displays. Common MR techniques like gestures or controllers might not be feasible or appropriate in mobile scenarios due to limited spaces, social acceptability or fatigue. To overcome such issues, previous work has already looked at using touch surfaces for interaction in MR [14] and we have compared the hand-tracking performance of several HMDs while operating touchscreens [12]. Additionally, touch devices such as tablets are already used in mobile knowledge worker scenarios.

In contrast to controllers, touchscreens are typically used as indirect input devices when interacting with virtual content beyond their physical display bounds. Therefore, the question arises how to efficiently interact with a large number of virtual screens using one, rather small touchscreen. Therefore, we proposed and evaluated a bimanual and a gaze-based technique that allow to efficiently switch between virtual screens while still providing the possibility for accurate input [2]. The bimanual technique included a mode switch by touching the bezel of the touchscreen with the non-dominant hand to switch from fast transitions between screens to precise input within a selected screen. In the gaze-based technique the transitions between screens were implicitly indicated by the gaze direction of the user and the touchscreen was solely used for precise input. Study results showed that the gaze-based technique was significantly faster and more usable. Therefore, eye-gaze, which is built-in in several

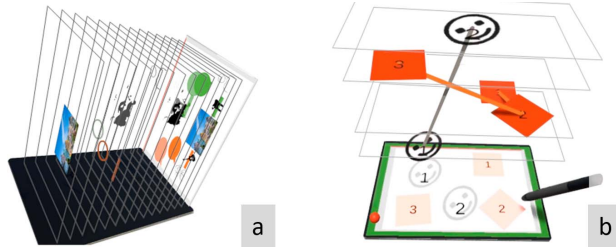


Figure 1: Presentation authoring techniques [1]: a) layered objects are presented slightly separated in depth which allows to efficiently reorder them b) creating animations with time of key-frames represented on z-axis above the screen and key-frame layers can be moved with a spatially tracked pen.

current head-mounted displays (HMDs), is a valuable addition to the set of interaction techniques for mobile knowledge work in VR.

On the one hand the large, three-dimensional virtual space can be used to display multiple screens containing resources relevant for the current tasks. On the other hand, it can be used to introduce interaction concepts not possible on a common screen that have the potential to facilitate knowledge work. For example, using the three-dimensional depth visualization provided by MR has shown to be efficient and usable [2]. We have applied this concept to a variety of concrete use-cases. The dependencies of nested functions in spreadsheet applications can be presented on different virtual layers above the actual screen [5], in animation authoring the timeline of key-frames can be mapped to the z-axis orthogonal to the screen [1] (Figure 1 b) and occlusion of layered objects can be handled by displaying the layers separated in depth (Figure 1 a). Our VR technique has been shown to outperform two state-of-the-art methods for reordering layered objects in presentation slides regarding performance and usability [1]. A main advantage here is the 3D view which allows the users to clearly discern between the layers and recognize which objects belong to which layers. This can be helpful for different kinds of applications involving layers or occlusions.

Mixed reality also provides new possibilities to interact with this three-dimensional space. For example, a touchscreen can be combined with a spatially tracked pen [5] [1]. Such a pen enables three-dimensional interaction with the virtual content but also precise input on the touch-surface.

3 OUTLOOK

To further evaluate the usability of working in MR and detect possible issues only arising after longer periods of use, we are currently conducting a study to evaluate the effects of long-term use of VR for knowledge work. In this study, participants are working in VR for one week full time doing their normal work to avoid bias through artificial tasks. We are collecting a wide range of data including subjective measures of productivity, usability, simulator sickness, visual fatigue, flow, presence, task load and anxiety and also objective measures like heart rate and break times. To get a baseline, all measures are also done while participants are working in a regular workplace setup for one week.

So far, we only looked at individual knowledge workers. Yet, collaborating with coworkers is an important aspect of today's working life. Therefore, in a next step, I will also explore interaction techniques to support collaborative knowledge work in MR.

4 CONCLUSION

So far, we have explored using eye-tracking, tablets and pens for certain knowledge worker tasks in VR and also how the three-dimensional display can enhance interactions. Our study results

indicated that such techniques are usable and can increase performance. Our ongoing study on long-term use will hopefully provide new insights into the advantages and challenges of using VR for knowledge work.

During the doctoral consortium, I welcome any feedback on the current process or the proposed next steps. I am wondering if I should also include augmented reality experiences in my next steps and if there are any aspects that were not mentioned that should be inquired in the scope of this thesis.

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